

SAJEEVAN VEERIAH

Mechatronics, Automation, Robotics and AI/ML Engineer

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PROFESSIONAL SUMMARY

Multidisciplinary full-stack engineer spanning physical systems, embedded firmware, controls, automation, robotics, CAD, data and applied AI/ML. Hands-on experience across industrial automation and smart factory systems, automotive software validation, emissions and compliance testing, manufacturing operations and end-to-end IoT deployments. Strong practical base across mechatronics, mechanical integration, electronics, electrical systems, embedded C/C++, Python, CAN, LoRaWAN, MQTT, InfluxDB, Grafana, ROS 2, SLAM and sensor fusion.

KEY ENGINEERING STRENGTHS

- End-to-end system integration from sensors, drives and process equipment to control logic, HMI, process visualisation, MES, telemetry and operational workflows.
- Automation and regulated manufacturing exposure across functional design, FAT, SAT, commissioning, qualification, diagnostics, data integrity and maintainability.
- Vehicle software and ADAS validation using structured test procedures, CAN analysis, fault isolation, regression testing and evidence-based defect reporting.
- Embedded and IoT implementation across ESP32 and STM32 concepts, sensor integration, firmware configuration, LoRaWAN commissioning, data pipelines and handover.
- Robotics and autonomy projects using ROS 2, SLAM, A* path planning, Kalman/EKF state estimation, LiDAR and IMU concepts, computer vision, PID control and simulation.

TECHNICAL SKILLS

- **Automation, Controls and SCADA:** PLC fundamentals, Siemens TIA Portal, WinCC, iFIX, PVI+, HMI, SCADA, MES, batch execution, FAT, SAT, commissioning, Modbus TCP/RTU, Profinet, industrial Ethernet and process troubleshooting.
- **Robotics and Autonomous Systems:** ROS 2, Gazebo, RViz, SLAM, A* path planning, Kalman filtering, EKF concepts, state estimation, sensor fusion, LiDAR, IMU, depth camera concepts, OpenCV, motion control and PID tuning.
- **Embedded Systems and Electronics:** ESP32, STM32 concepts, embedded C/C++, Python, FreeRTOS concepts, UART, I2C, SPI, CAN, ADC, PWM, sensor integration, actuator control, board-level debugging, Altium, KiCad and signal conditioning.
- **Software, Data and AI/ML:** Python, C, C++, MATLAB, Simulink, JavaScript, Node.js, REST APIs, Git, Linux, JIRA, data analysis, test automation, InfluxDB, Grafana, applied ML fundamentals, statistical modelling, feature extraction and anomaly detection concepts.
- **CAD, Mechanical and Electrical:** SolidWorks, Fusion 360, part and assembly modelling, drawing interpretation, mechanical integration, FEA exposure, 3D printing, control schematics, VFD exposure, sensors, actuators, pneumatics and hydraulics.
- **Automotive, Quality and Compliance:** Vector CANoe, CANalyzer, CAN bus, CAN FD, ADAS testing, OTA regression, AVL/emissions test systems, ADR, EURO, ISO, IEC, AS/NZS, ITPs, MDRs, FMEA, QA records and traceability.

PROFESSIONAL EXPERIENCE

Automation Engineer | JAG Process Solutions Pty Ltd | Brunswick, VIC | Jan 2026 to Jun 2026

- Worked on automation and smart factory systems for pharmaceutical, biotech, food and process clients where reliability, traceability and validation discipline are central requirements.
- Supported control logic, HMIs, process visualisation, MES, batch execution and production reporting data across automated plants, skids and package units.
- Contributed to requirements, functional design, FAT, SAT, commissioning, qualification and troubleshooting activities.
- Connected field devices, sensors, drives and process equipment with HMI, control logic, production data and operational workflows while considering diagnostics, safety and maintainability.

Product Development Test Engineer | Ford Motor Company via Invenio | Little River, VIC | Oct 2025 to Jan 2026

- Worked in the Feature Software Integration and Validation team on vehicle software integration and ADAS validation for T6 Ranger and Everest programmes.
- Executed feature vehicle, breadboard and regression testing for readiness, running changes, OTA updates and sign-off evidence.
- Handled vehicle instrumentation and structured test drives in controlled and real-world conditions using corporate test procedures.
- Used Vector CANoe, CANalyzer and proprietary diagnostic tools to capture and analyse CAN bus data for fault isolation and verification.

Technical Assistant | ABMARC | South Geelong, VIC | Jul 2024 to Aug 2025

- Supported vehicle emissions and compliance testing against Australian Design Rules and EURO standards across a range of vehicle platforms.
- Operated emissions testing equipment, instrumentation and data acquisition systems with a focus on repeatable, auditable and technically defensible results.
- Prepared compliance, QA and regulatory documentation for certification, audits and reporting while analysing emissions and test data to identify deviations.

PROFESSIONAL EXPERIENCE CONTINUED

Consultant Engineer - Internet of Things | DuxTel Pty Ltd | Greater Geelong, VIC | May 2024 to Aug 2024

- Designed and deployed end-to-end IoT solutions linking sensors, embedded devices, gateways, cloud platforms and data pipelines.
- Built real-time data ingestion, monitoring and visualisation using InfluxDB and Grafana, with documentation and post-deployment support for handover.

IoT Projects Administrator | DuxTel Pty Ltd | Greater Geelong, VIC | Feb 2024 to May 2024

- Configured and integrated IoT devices into ChirpStack-based LoRaWAN networks for low-power, wide-area deployments.
- Developed an ESP32-based GPS tracking system with environmental sensing, firmware configuration, device provisioning, testing and deployment workflows.

Production Worker and Cellar Hand | IDL | Moorabool, VIC | Aug 2022 to Feb 2024

- Ran and monitored production line equipment, maintained throughput and quality, completed checks and reported faults in a scheduled food and beverage production environment.

Production Line Operator | Carbon Revolution | Waurin Ponds, VIC | Sep 2021 to Dec 2021

- Set up and operated manufacturing equipment for high-performance carbon fibre wheels while maintaining quality, safety, maintenance and first-level troubleshooting requirements.

Undergraduate Engineer | Thornton Engineering Australia Pty Ltd | Corio, VIC | Mar 2018 to Mar 2020

- Supported engineering and QA for structural steel fabrication delivered to Australian Standards and client specifications.
- Prepared Inspection Test Plans, Manufacturer Data Reports and quality records, and reviewed drawings for inspection planning, material traceability and compliance.

SELECTED ENGINEERING PROJECTS

- **Autonomous Navigation and ROS 2 Robotics:** Built robotics software concepts using ROS 2, A* path planning, SLAM, Kalman/EKF-based state estimation, odometry and IMU fusion, simulation and modular control nodes.
- **ESP32 Clinical Ataxia Assessment Concept:** Developed an embedded concept using ESP32, IMU, ToF, Hall and magnetometer sensing, firmware logic, data acquisition and analysis workflows for measurement support.
- **IoT GPS and Environmental Monitoring Platform:** Developed ESP32 GPS and environmental sensing with LoRaWAN/MQTT telemetry, gateway/device configuration, data ingestion, InfluxDB/Grafana dashboards and deployment documentation.
- **Smart Factory and Process Visualisation Exposure:** Worked across HMI/process visualisation, MES and batch-system concepts where operator usability, traceability, diagnostics and maintainability must be engineered together.

EDUCATION, MEMBERSHIPS, TRAINING AND LANGUAGES

- Bachelor of Mechatronics Engineering (Honours), Deakin University, VIC, Mar 2018 to Jul 2025.
- Higher National Diploma, Mechatronics, Robotics and Automation Engineering, Cardiff Metropolitan University, 2014 to 2016.
- Professional membership: Member, Engineers Australia.
- Training: Lean Six Sigma Foundation, JIRA and Agile, KAIZEN.
- Languages: English, Tamil and Sinhala.